

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 09

DATE : 22-09-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

www.allen.ac.in

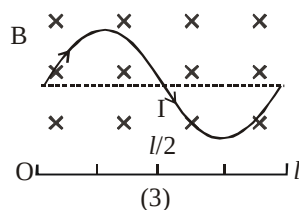
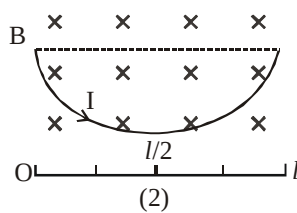
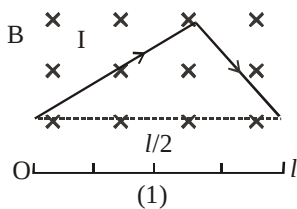
Your Target is to secure Good Rank in NEET 2020

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

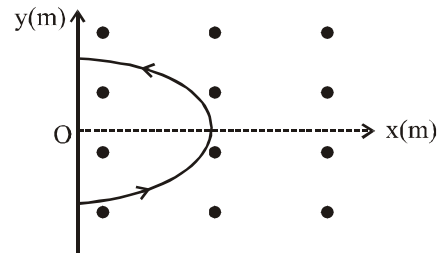
1. Three conductors 1, 2 and 3 each carrying the same current I are placed in a uniform magnetic field B , as shown in fig.



The force experienced by conductors 1, 2 and 3 are F_1 , F_2 and F_3 , respectively.

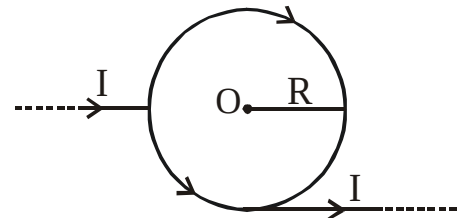
- (1) $F_3 > F_2 > F_1$
- (2) $F_1 \neq 0$; $F_2 \neq 0$; $F_3 = 0$
- (3) F_1 acts upwards, F_2 acts downwards; $F_3 = 0$.
- (4) All experience the same force in same direction

2. A wire carrying a current 3A is bent in the form of a parabola $y^2 = 4 - x$ as shown in figure, where x and y are in metre. The wire is placed in a uniform magnetic field $\vec{B} = 5\hat{k}$ tesla. The force acting on the wire is :-



- (1) $60\hat{i}$ N
- (2) $-60\hat{i}$ N
- (3) $30\hat{i}$ N
- (4) $-30\hat{i}$ N

3. In the arrangement of wires as shown in fig. the magnitude of B at the center O is

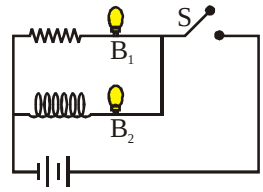


- (1) $\frac{\mu_0 I}{4R} - \frac{\mu_0 I}{4\pi R}$ (inwards)
- (2) $\frac{\mu_0 I}{4R} - \frac{\mu_0 I}{2\pi R}$ (inwards)
- (3) $\frac{\mu_0 I}{4\pi R}$ (outwards)
- (4) $\frac{\mu_0 I}{2\pi R}$

4. An electron is projected at an angle θ with a uniform magnetic field. If the pitch of the helical path is equal to the its radius, then the angle of projection is

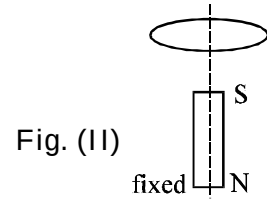
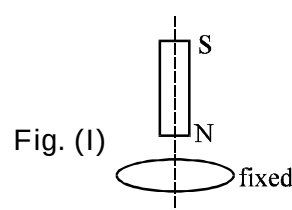
- (1) $\tan^{-1}\pi$
- (2) $\tan^{-1}2\pi$
- (3) $\cot^{-1}\pi$
- (4) $\cot^{-1}2\pi$

SPACE FOR ROUGH WORK

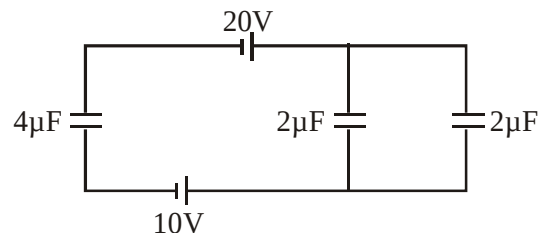
5. A varying current in a coil changes from 10 amp to zero in 0.5 sec. If average EMF is induced in the coil is 220 volts, the self inductance of coil is :-
 (1) 5 H (2) 10 H
 (3) 11 H (4) 12 H
6. A charge 'q' moves in a region where electric field and magnetic field both exist, then force on it is
 (1) $q(\vec{v} \times \vec{B})$ (2) $q\vec{E} + q(\vec{B} \times \vec{v})$
 (3) $q\vec{B} + q(\vec{E} \times \vec{v})$ (4) $q\vec{E} + q(\vec{v} \times \vec{B})$
7. An electron and a proton have equal kinetic energies. They enter in a magnetic field perpendicularly then
 (1) Both will follow a circular path with same radius
 (2) Both will follow a helical path
 (3) Both will follow a parabolic path
 (4) All the statements are false
8. A vibration magnetometer consists of two identical bar magnets placed one over the other such that they are perpendicular and bisect each other. The time period of oscillation in a horizontal magnetic field is $2^{5/4}$ seconds. One of the magnets is removed and if the other magnet oscillates in the same field, then the time period in seconds is :
 (1) $2^{1/4}$ (2) $2^{1/2}$
 (3) 2 (4) $2^{3/4}$
9. A current I flows along the length of an infinite long, straight thin walled pipe. Then :-
 (1) The magnetic field at all points inside the pipe is the same, but not zero
 (2) The magnetic field is zero only on the axis of the pipe
 (3) The magnetic field is different at different points inside the pipe
 (4) The magnetic field at any point inside the pipe is zero
10. The adjoining figure shows two bulbs B_1 and B_2 resistor R and an inductor L. When the switch S is turned on :-

 (1) Both B_1 and B_2 glow promptly
 (2) Both B_1 and B_2 glow with some delay
 (3) B_1 glow promptly but B_2 with some delay
 (4) B_2 glow promptly but B_1 with some delay
11. A coil having 500 square loops each of side 10 cm is placed normal to a magnetic flux which increases at the rate of 1.0 Tesla/second. The induced e.m.f. in volts is :-
 (1) 0.1 (2) 0.5 (3) 1 (4) 5
12. The plane at which the Earth's magnetic field is horizontal is :-
 (1) Plane through magnetic meridian
 (2) Plane through magnetic equator
 (3) Plane through magnetic poles
 (4) It is not horizontal anywhere

SPACE FOR ROUGH WORK

13. The Earth's magnetic field at a given point is $0.5 \times 10^{-5} \text{ Wb/m}^2$. This field is to be neutralized by magnetic induction at the centre of a circular loop of radius 5 cm. The current required to be flown in the loop is nearly :-
 (1) 0.2 A (2) 0.4 A (3) 4 A (4) 40 A
14. The materials suitable for making electromagnets should have :-
 (1) High retentivity and high coercivity
 (2) Low retentivity and low coercivity
 (3) High retentivity and low coercivity
 (4) Low retentivity and high coercivity
15. The universal property among all substances is:-
 (1) Diamagnetism
 (2) Paramagnetism
 (3) Ferromagnetism
 (4) non-magnetism
16. A magnetic substance has susceptibility – 0.05 at 300 K. The magnetic susceptibility of the substance at 600 K will be :-
 (1) –0.025 (2) –0.1
 (3) –0.05 (4) –0.0125
17. A bar magnet is placed inside a non-uniform magnetic field. It may experience :-
 (1) force and torque
 (2) force but not torque
 (3) torque but not force
 (4) Neither force nor torque
18. A vertical bar magnet is dropped from position on the axis of a fixed metallic coil as shown in fig - I. In fig. II the magnet is fixed and horizontal coil is dropped. The acceleration of the magnet and coil are a_1 and a_2 respectively then



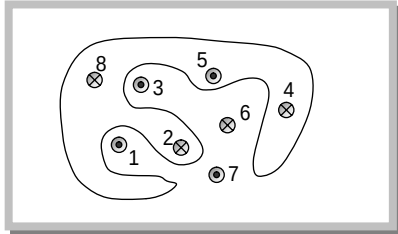
- (1) $a_1 > g, a_2 > g$ (2) $a_1 > g, a_2 < g$
 (3) $a_1 < g, a_2 < g$ (4) $a_1 < g, a_2 > g$
19. A proton (mass m and charge $+e$) and an α -particle (mass $4m$ and charge $+2e$) are projected with the same kinetic energy at right angles to the uniform magnetic field. Which one of the following statements will be true ?
 (1) The α -particle will be bent in a circular path with a small radius that for the proton
 (2) The radius of the path of the α -particle will be greater than that of the proton
 (3) The α -particle and the proton will be bent in a circular path with the same radius
 (4) The α -particle and the proton will go through the field in a straight line
20. In the circuit shown in Fig., charge stored in $4\mu\text{F}$ capacitor is



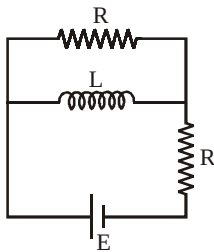
- (1) $20 \mu\text{C}$ (2) $10 \mu\text{C}$
 (3) $40 \mu\text{C}$ (4) $120 \mu\text{C}$

SPACE FOR ROUGH WORK

21. Eight wires cut the page perpendicularly at the points shown. Each wire carries current i_0 . Odd currents are out of the page and even currents into the page. The line integral $\int \vec{B} \cdot d\vec{l}$ along the loop is



- (1) $\mu_0 i_0$ (2) $2\mu_0 i_0$
(3) 0 (4) $3\mu_0 i_0$
22. A parallel plate capacitor has capacitance of 10^{-12} F. The separation of the plates is doubled and wax is inserted between them which increases the capacitance to 2×10^{-12} F. The dielectric constant of wax is :
- (1) 2 (2) 3 (3) 4.0 (4) 8.0
23. The figure shows an L-R circuit, the time constant for the circuit is



- (1) $\frac{L}{2R}$ (2) $\frac{2L}{R}$
(3) $\frac{2R}{L}$ (4) $\frac{R}{2L}$

24. A conducting circular loop is placed in a uniform magnetic field of inducing B tesla with its plane normal to the field. Now, radius of the loop starts shrinking at the rate (dr/dt) . Then the induced e.m.f. at the instant when the radius is r is :-

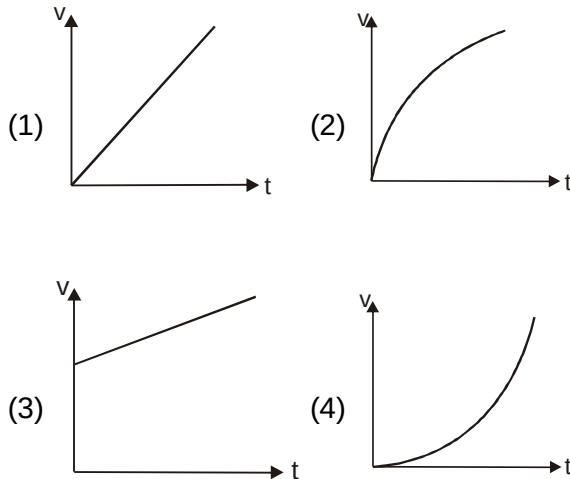
- (1) $\pi r B \left(\frac{dr}{dt} \right)$ (2) $2\pi r B \left(\frac{dr}{dt} \right)$
(3) $\pi r^2 B \left(\frac{dr}{dt} \right)$ (4) $\frac{\pi B r^2}{2} \left(\frac{dr}{dt} \right)$

25. If for a particle position $x = \kappa t^2$ then :-
(1) velocity is constant
(2) acceleration is constant
(3) acceleration is variable
(4) None of these
26. The period of revolution of planet A around the sun is 8 times that of B. The distance of A from the sun is n times greater than that the distance of B from the sun. The value of n is :-
(1) 4 (2) 8
(3) 12 (4) 16
27. The energy required to shift a satellite from orbital radius r to orbital radius 2r is E. What energy will be required to shift the satellite from orbital radius 2r to orbital radius 3r?

- (1) E (2) $\frac{E}{2}$
(3) $\frac{E}{3}$ (4) $\frac{E}{4}$

SPACE FOR ROUGH WORK

28. A particle starts from rest and moves along a straight line with constant acceleration. The variation of velocity v with time is :



29. A body is moving according to the equation $x = at + bt^2 - ct^3$. Then its instantaneous speed is given by :-
 (1) $a + 2b + 3ct$
 (2) $a + 2bt - 3ct^2$
 (3) $2b - 6ct$
 (4) none of these
30. A scooter is going towards east at 10 m s^{-1} turns right through an angle of 90° . If the speed of the scooter remains unchanged in taking this turn, the change in the velocity of the scooter is :-
 (1) 20.0 ms^{-1} in south-western direction
 (2) zero
 (3) 10.0 ms^{-1} in southern direction
 (4) 14.14 ms^{-1} in south-western direction

31. A particle is thrown vertically upwards from the surface of earth and it reaches to a maximum height equal to the radius of earth. The ratio of the velocity of projection to the escape velocity on surface of earth is :-

(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) $\frac{1}{2\sqrt{2}}$

32. A satellite which is geostationary in a particular orbit is taken to another orbit. Its distance from the center of earth in the new orbit is 2 times that of the earlier orbit. The time period in the new orbit is :-

(1) 48 hrs. (2) $48\sqrt{2}$ hrs.
 (3) 24 hrs. (4) $24\sqrt{2}$ hrs.

33. A planet has twice the density of earth but the acceleration due to gravity on its surface is exactly the same as that on the surface of earth. Its radius in terms of earth's radius R will be :-

(1) $\frac{R}{4}$ (2) $\frac{R}{2}$ (3) $\frac{R}{3}$ (4) $\frac{R}{8}$

34. Mark the incorrect option :-

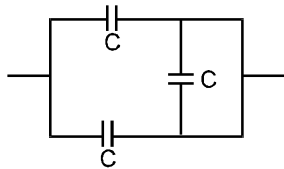
- (1) When an uncharged capacitor is connected across the battery, just after the connections has been made, it behaves as short circuit.
 (2) When capacitor is fully charged it behave as short circuit
 (3) When capacitor is fully charged it is behaving as open circuit
 (4) (1) & (3) both

SPACE FOR ROUGH WORK

35. Mark the correct option :-

- (1) When an inductor is connected across the battery, just after the connections has been made, it behaves as short circuit.
- (2) When an inductor carries steady current , it behave as short circuit
- (3) When an inductor carries steady current , it is behaving as open circuit
- (4) (1) & (3) both

36. The equivalent capacitance of the combination shown in Figure is.



- (1) C
- (2) 2C
- (3) $\frac{3}{2}C$
- (4) $\frac{C}{2}$

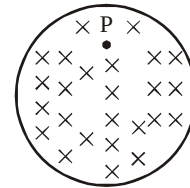
37. A capacitor stores q charge when connected with a battery. If the gap between plates is filled with a dielectric, an additional charge $3q$ flows through the battery. The dielectric constant of material filled is :-

- (1) 2
- (2) 1
- (3) 4
- (4) 3

38. A circular current loop of magnetic moment M is in an arbitrary orientation in an external magnetic field B . The work done to rotate the loop by 30° about an axis perpendicular to its plane is

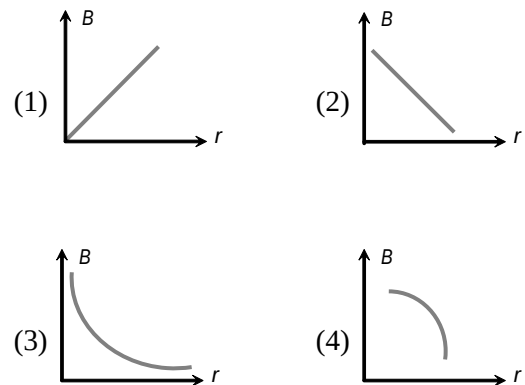
- (1) MB
- (2) $\sqrt{3}\frac{MB}{2}$
- (3) $\frac{MB}{2}$
- (4) zero

39. A uniform but increasing with time magnetic field exists in a cylindrical region. The direction of force on an electron at P is



- (1) towards right
- (2) towards left
- (3) into the plane of paper
- (4) out of the plane of paper

40. Which of the following graphs shows the variation of magnetic induction B with distance r from a long wire carrying current



SPACE FOR ROUGH WORK

41. A bar magnet of magnetic moment m and moment of inertia I (about centre, perpendicular to length) is cut into two equal pieces, perpendicular to length. Let T be the period of oscillations of the original magnet about an axis through the mid point, perpendicular to length, in a magnetic field B . What would be the similar period T' for each piece?

- (1) T (2) $T/2$
(3) $T/4$ (4) $2T$

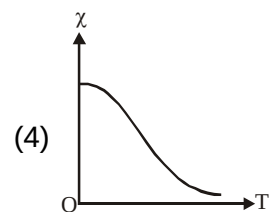
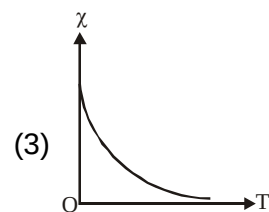
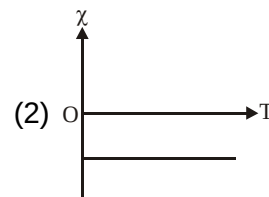
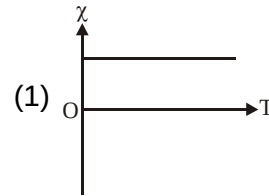
42. A long solenoid has 1000 turns per metre and carries a current of 1 A. It has a soft iron core of $\mu_r = 1000$. The core is heated beyond the Curie temperature, T_c .

- (1) The H field in the solenoid is (nearly) unchanged but the B field decreases drastically
(2) The H and B fields in the solenoid are nearly unchanged.
(3) The magnetisation in the core reverses direction.
(4) The magnetisation in the core diminishes by a factor of about 10.

43. The primary origin(s) of magnetism lies in

- (1) atomic currents.
(2) Intrinsic spin of electron
(3) polar nature of molecules.
(4) Both (1) and (2)

44. The variation of magnetic susceptibility (χ) with temperature for a diamagnetic substance is best represented by :-

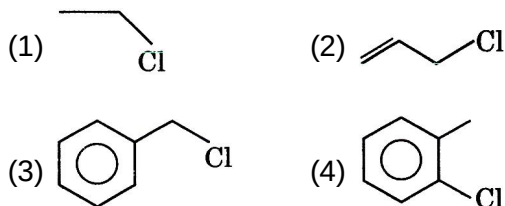


45. A body covered a distance of 5 m along a semicircular path. The ratio of distance to displacement is :-

- (1) 11 : 7 (2) 12 : 5
(3) 8 : 3 (4) 7 : 5

SPACE FOR ROUGH WORK

46. In which of the following, replacement of Cl^- is most difficult?



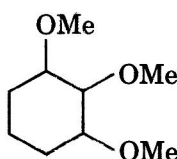
47. $\text{ClCH}_2\text{CHCl}_2 \xrightarrow{\text{alc. KOH}} \text{A}$

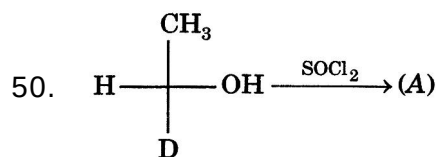
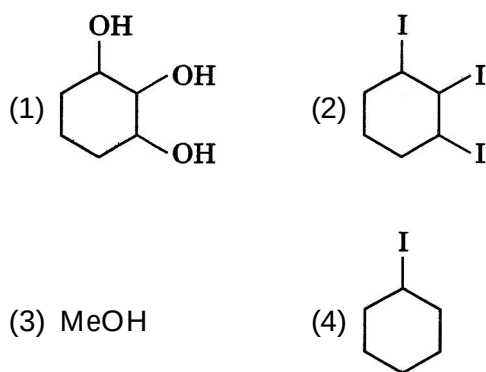
A is :

- (1) $\text{CHCl} = \text{CHCl}$
(2) $\text{CH}_2 = \text{CCl}_2$
(3) $\text{HO}-\text{CH}_2-\text{CH}=\text{O}$
(4) ClCH_2CHO

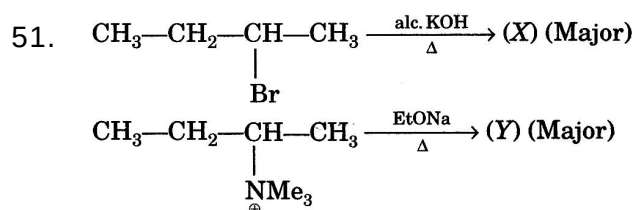
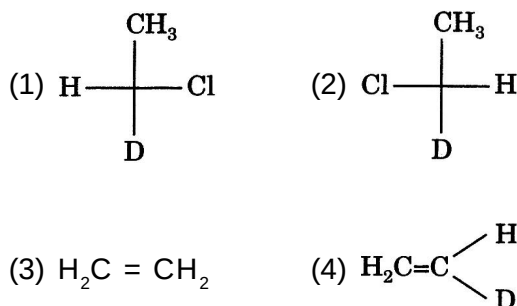
48. True statement regarding $\text{S}_\text{N}1$ reaction is :

- (1) perfect racemisation is observed
(2) only Walden inversion is observed
(3) total retention of configuration is observed
(4) none of these

49.  $\xrightarrow[\Delta]{\text{HI excess}}$ major product is :



Major product (A) is :-



Product (X) and (Y) respectively are :

- (1) 1-butene, trans-2-butene
(2) 1-butene, cis-2-butene
(3) cis-2-butene, 1-butene
(4) trans-2-butene, 1-butene

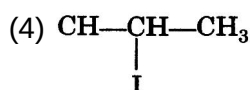
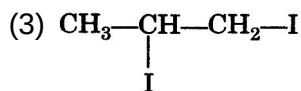
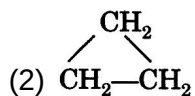
52. Which of the alkyl halides will undergo $\text{S}_\text{N}1$ reaction at a fastest rate?

- (1) $\text{Cl}-\text{CH}_2-\text{CN}$
(2) $\text{Cl}-\text{CH}_2-\text{NO}_2$
(3) $\text{Cl}-\text{CH}_2-\text{OMe}$
(4) $\text{Cl}-\text{CH}_2-\text{CH}_3$

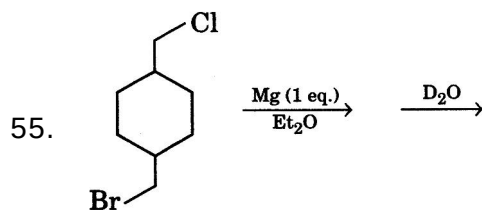
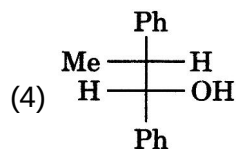
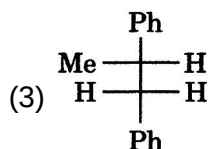
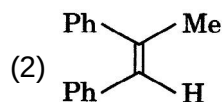
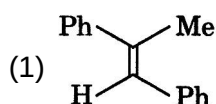
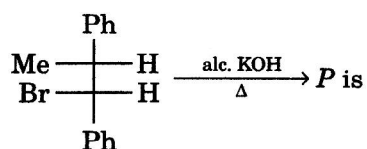
SPACE FOR ROUGH WORK

53. $\text{CH}_2=\text{CH}-\text{CH}_2-\text{I} \xrightarrow[\Delta]{\text{HI(excess)}}$ major product.

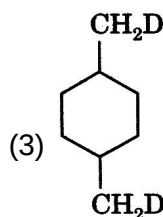
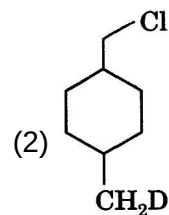
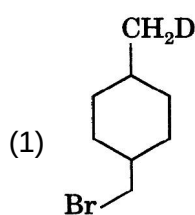
(1) $\text{I}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{I}$



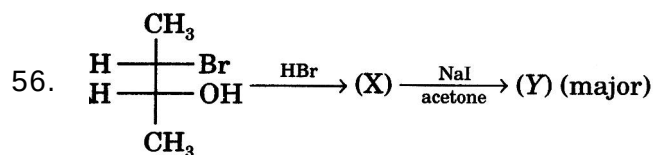
54. The organic Product (P) of the reaction.



major product is :-



(4) none of these

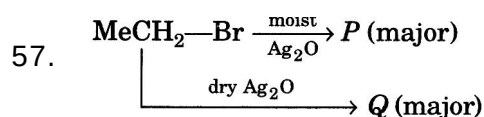


(1) cis-2-butene

(2) trans-2-butene

(3) 1-butene

(4) isobutene



(1) P and Q are MeCH_2OAg

(2) P and Q are MeCH_2OH

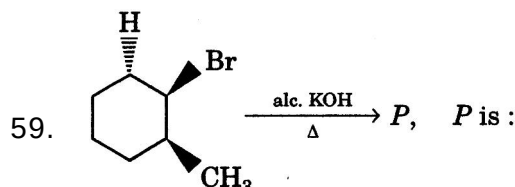
(2) P and Q are $\text{MeCH}_2\text{OCH}_2\text{Me}$

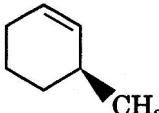
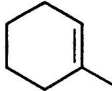
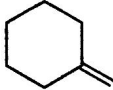
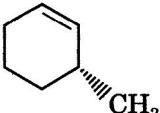
(4) P is MeCH_2OH and Q is $\text{MeCH}_2\text{OCH}_2\text{Me}$

SPACE FOR ROUGH WORK

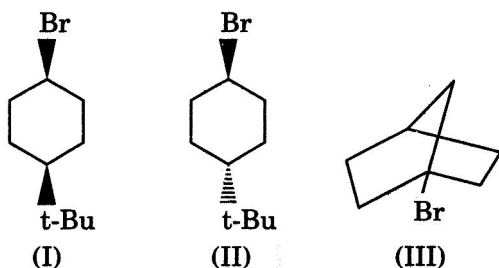
58. Which of the following nucleophile will show minimum reactivity towards S_N2 reaction?

- (1) Me_3CO^- (2) MeO^-
(3) EtO^- (4) Me_2CHO^-



- (1)  (2) 
(3)  (4) 

60. The rate of elimination (using EtONa) of :-



- (1) I > II > III (2) I > III > II
(3) II > I > III (4) III > I > II

61. $(\text{CH}_3)_3\text{CMgCl}$ on reaction with D_2O produces :

- (1) $(\text{CH}_3)_3\text{CD}$ (2) $(\text{CH}_3)_3\text{OD}$
(3) $(\text{CD}_3)_3\text{CD}$ (4) $(\text{CD}_3)_3\text{OD}$

62. The order of reactivities of the following alkyl halides for a S_N2 reaction is :

- (1) $\text{RF} > \text{RCl} > \text{RBr} > \text{RI}$
(2) $\text{RF} > \text{RBr} > \text{RCl} > \text{RI}$
(3) $\text{RCl} > \text{RBr} > \text{RF} > \text{RI}$
(4) $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$

63. On monochlorination of 2-methylbutane, the total number of chiral compounds formed is :

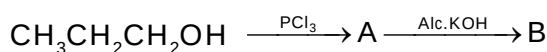
- (1) 2 (2) 4 (3) 6 (4) 8

64. The reagent(s) for the following conversion is / are :

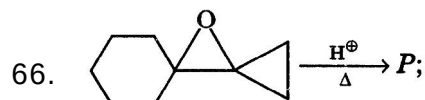


- (1) alcoholic KOH
(2) aqueous KOH
(3) aqueous KOH followed by NaNH_2
(4) $\text{Zn}/\text{CH}_3\text{OH}$

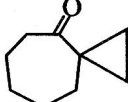
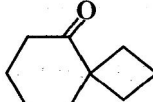
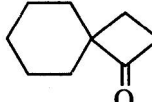
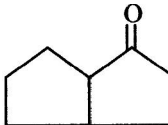
65. The compound B formed in the following sequences of reactions is



- (1) Propyne (2) Propene
(3) Propanol (4) Propane

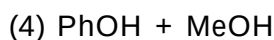
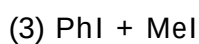
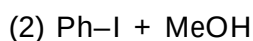
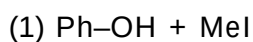
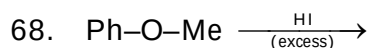
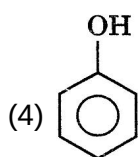
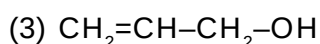
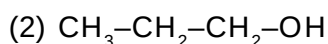
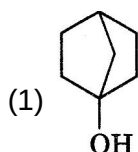


Major product of reaction is :-

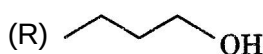
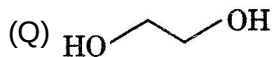
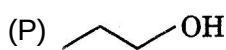
- (1)  (2) 
(3)  (4) 

SPACE FOR ROUGH WORK

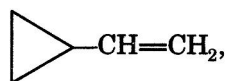
67. Which of the following can give immediate turbidity on treatment Lucas Reagent?



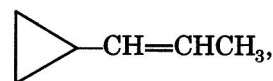
69. The correct increasing order of boiling point of the following alcohols is :-



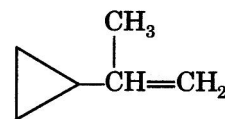
70. Rate of hydration of :



(I)

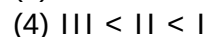
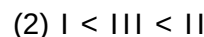


(II)

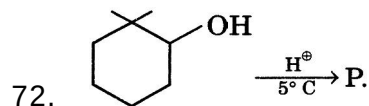
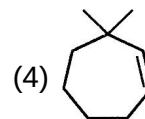
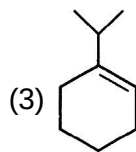
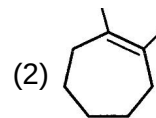
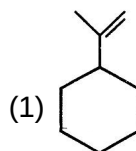
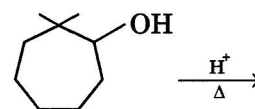


(III)

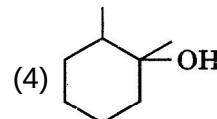
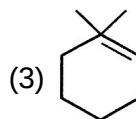
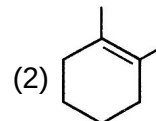
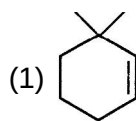
will be in order :



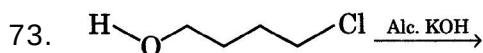
71. Which one of the following can not be the product during dehydration of following alcohol?



The product P is :-

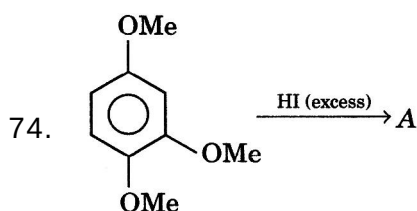


SPACE FOR ROUGH WORK



Major product is :

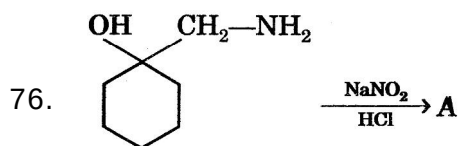
- (1) 
- (2) 
- (3) 
- (4) 



A is :-

- (1) 
- (2) 
- (3) 
- (4) MeOH

75. Which of the following is the correct order of decreasing S_N2 reactivity ?
- (1) $RCH_2X > R_3CX > R_2CHX$
 - (2) $RCH_2X > R_2CHX > R_3CX$
 - (3) $R_3CX > R_2CHX > RCH_2X$
 - (4) $R_2CHX > R_3CX > RCH_2X$
- (X = a halogen)

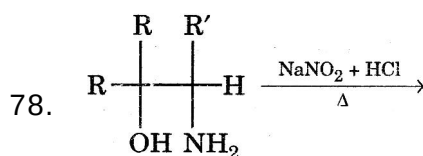


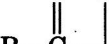
A is :-

- (1) 
- (2) 
- (3) 
- (4) 

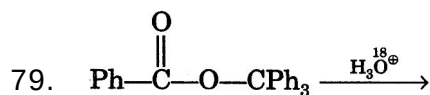
77. Which of the following alcohols cannot be prepared from an alkene?

- (1)  (2) 
- (3)  (4) 

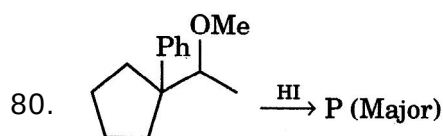
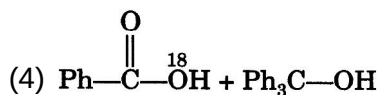
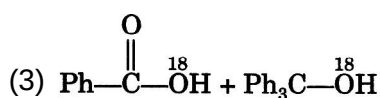
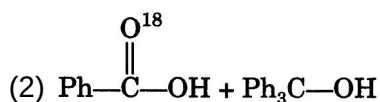
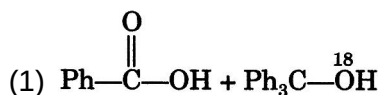


- (1) 
- (2) 
- (3) 
- (4) 

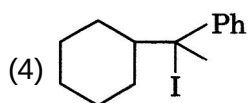
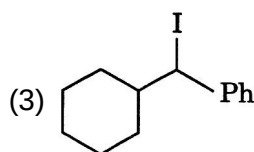
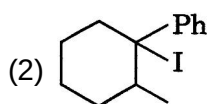
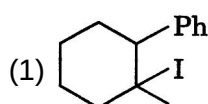
SPACE FOR ROUGH WORK



Product of this reaction is :-



'P' is :-



81. The compound which reacts fastest with Lucas reagent at room temperature is :

- (1) butan-1-ol
- (2) butan-2-ol
- (3) 2-methylpropan-1-ol
- (4) 2-methylpropan-2-ol

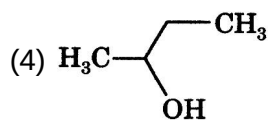
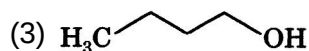
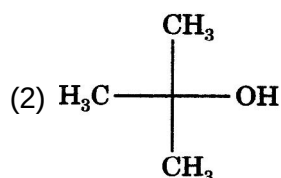
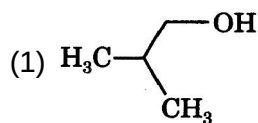
82. HBr reacts fastest with :

- (1) 2-methylpropan-2-ol
- (2) propan-1-ol
- (3) propan-2-ol
- (4) 2-methylpropan-1-ol

83. In $\text{CH}_3\text{CH}_2\text{OH}$, the bond that undergoes heterolytic cleavage most readily is :

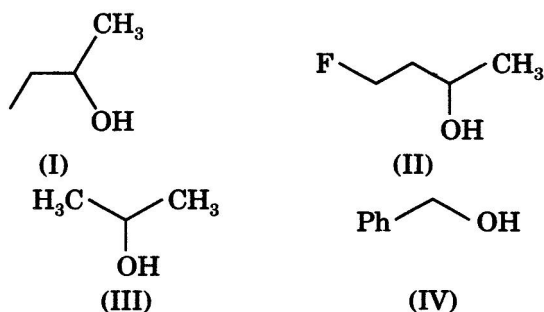
- (1) C—C
- (2) C—O
- (3) C—H
- (4) O—H

84. The compound which gives the most stable carbonium ion on dehydration is :



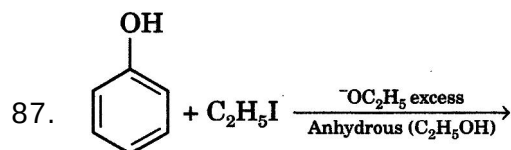
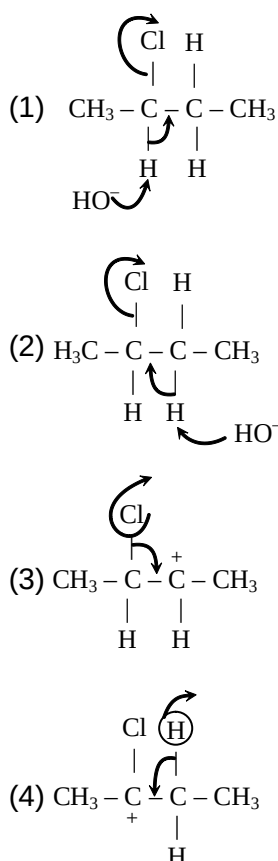
SPACE FOR ROUGH WORK

85. The order of reactivity of the following alcohols towards conc. HCl is :



- (1) I > II > III > IV (2) I > III > II > IV
(3) IV > III > II > I (4) IV > III > I > II

86. For a compound undergoing E₂ reaction which step is correct -



- (1) C₆H₅OC₂H₅
(2) C₂H₅OC₂H₅
(3) C₆H₅OC₆H₅
(4) C₆H₅I

88. The best method to prepare cyclohexene from cyclohexanol is by using:

- (1) conc. HCl + ZnCl₂ (Lucas Reagent)
(2) conc. H₃PO₄ (Dehydrating Agent)
(3) conc. HBr
(4) conc. HCl

89. In the reaction  $\xrightarrow{\text{HBr}}$ the products are :

- (1) Br--OCH₃ and H₂
(2) -Br and CH₃Br
(3) -Br and CH₃OH
(4) -OH and CH₃Br

90. Which of the following are electrophile?

- (1) CH₃⁻ (2) OH[⊖]
(3) NO₂⁺ (4) H₂O

SPACE FOR ROUGH WORK

91. Which among the following scientists had no contribution in the development of the double helix model for the structure of DNA :-

- (1) Watson
- (2) Wilkins
- (3) Franklin
- (4) Singer

92. In a DNA strand the nucleotides of a chain are linked together by :-

- (1) Glycosidic bonds
- (2) Peptide bond
- (3) Phosphodiester bond
- (4) Hydrogen bond

93. With regards to mature m-RNA in eukaryotes :-

- (1) Exons and introns do not appear in the mature m-RNA.
- (2) Exons appear but introns do not appear in the mature m-RNA.
- (3) Introns appear but exons do not appear in the mature m-RNA
- (4) Both exons and introns appear in mature m-RNA.

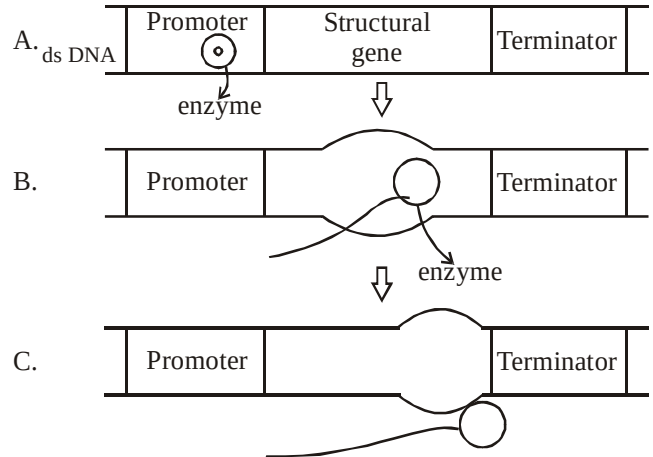
94. Read the following statement :-

- (i) r-RNA is a constituent component of ribosomes.
- (ii) The amino acid attaches to the t-RNA at its 5' end.
- (iii) m-RNA is a carrier (messenger) of genetic information to synthesising polypeptides.
- (iv) In eukaryotes RNA polymerase II enzyme is used to formation of t-RNA from DNA.

How many of the above is/are true statement.

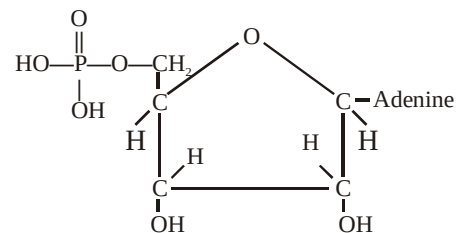
- (1) Two
- (2) Three
- (3) Four
- (4) One

95. The figure below shows three steps (A, B, C) of transcription. Select the option giving correct identification together with what it representation :-



(1)	B	Termination-with the help of realising factor
(2)	A	Initiation-with the help of Rho(ρ) factor
(3)	B	Elongation-with the help of DNA polymerase enzyme.
(4)	A	Initiation-with the help of sigma (σ) factor

96. Following structure is :-



- (1) Deoxyribonucleotide
- (2) Ribonucleotide
- (3) Deoxyribonucleoside
- (4) Ribonucleoside

97. Which of the following is related to degeneracy of genetic code ?

- (1) Some amino acids coded by only one codon.
- (2) One codon coding for only one amino acid.
- (3) One amino acid coded by more than one codon.
- (4) One codon coding for many amino acids.

98. Which of the following radio isotopes are used by Hershey and Chase to label DNA and Protein respectively :-

- (1) P^{32} , S^{35} (2) S^{35} , P^{32}
(3) N^{15} , P^{32} (4) P^{32} , N^{15}

99. Which of the following is incorrect according to Griffith experiments :-

- (A) S-Strain → Inject into mice → mice live
(B) R-Strain → Inject into mice → mice live
(C) Heat-killed S-Strain → Inject into mice → mice live
(D) Heat killed S-Strain + R-Strain → Inject into mice → mice died

Option :

- (1) A and D (2) Only A
(3) B and C (4) Only B

100. Find out the correct match from the following table.

	Column-I	Column-II	Column-III
(i)	UAA	Tyrosine	Stop codon
(ii)	UUU	Phenyl alanine	Start codon
(iii)	AUG	Methionine	Start codon
(iv)	UGA	Valine	Stop codon

Options :-

- (1) i and iv
(2) ii and iii
(3) iii only
(4) ii only

101. A molecule that can act as genetic material must not have following properties :-

- (1) It should be chemically and structurally highly active.
(2) It should be able to generate its replica.
(3) It should provide the scope for slow changes.
(4) It should be structurally more stable.

102. Which of the following RNA is known as adapter RNA ?

- (1) m-RNA (2) t-RNA
(3) r-RNA (4) Sn-RNA

103. Which of the following is not a salient features of genetic code ?

- (1) Codon is ambiguous and specific
(2) Codon is nearly universal
(3) Codon is read in mRNA in a contiguous fashion
(4) Codon is degenerate

104. Which of the following statement is correct ?

- (1) hn RNA is subjected to a process called splicing where the exons are removed
(2) In capping an unusual nucleotides is added to the 3' end of hn RNA
(3) Splicing represents the dominance of DNA world
(4) In tailing, adenylated residues are added in a template independent manner

105. When chromatins are seen under electron microscope the "beads-on-string". Structure appear thin structure are :-

- (1) Spliceosome
(2) Kinetochore
(3) Nucleosome
(4) Nucleoside

106. If ds DNA molecule contain 120 adenine and 120 cytosine bases. The total number of nucleotides in this DNA fragment is :-

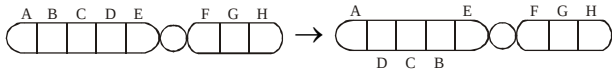
- (1) 120 (2) 240
(3) 60 (4) 480

107. Experiments using N^{15} to confirm semiconservative replication of DNA were carried out by:-

- (1) Messelson & Stahl
(2) Hershey & Chase
(3) Watson & Crick
(4) Taylor

108. DNA dependent, DNA polymerase enzyme is used in the :-

- (1) Transcription
(2) Splicing of Hn-RNA
(3) DNA – replication
(4) Synthesis of RNA – primer

109. Which of the following is not present in DNA :-
 (1) Phosphate
 (2) Deoxyribose sugar
 (3) Ribose sugar
 (4) Purine N base
110. Aminoacyl synthetase enzyme associates with :-
 (1) With small subunit of ribosome
 (2) TΨC loop of tRNA
 (3) 5'-end of mRNA
 (4) DHU loop of tRNA
111. Which is correct?
 (1) RNA is genetic material of bacteria
 (2) RNA is genetic material of all viruses
 (3) DNA is genetic material of some organism
 (4) Some viruses have RNA as genetic material
112. According to double helix model, the backbone of DNA is formed by :
 (1) Sugar and phosphate
 (2) Sugar and N₂-base
 (3) N₂ base, sugar, phosphate
 (4) Phosphate and N₂ base
113. In prokaryote which of the following is used as a ribozyme :-
 (1) 23s r RNA
 (2) 18s r RNA
 (3) 16s r RNA
 (4) 5s r RNA
114. Which one of the following is a real inducer of lac operon.
 (1) Lactose
 (2) IPTG
 (3) Allolactose
 (4) β - galactosidase
115. In lac operon y gene code an enzyme which is responsible for :-
 (1) Hydrolysis of lactose
 (2) Increases permeability of the cell to lactose
 (3) Transfer of acetyl group
 (4) Repression of lac operon
116. Given below is a representation of kind of chromosomal mutation. What is the kind of mutation represented ?

- (1) Deletion
 (2) Duplication
 (3) Paracentric inversion
 (4) Pericentric inversion
117. Which of following statement is true for lac operon in E-coli.
 (a) lac operon consists of one regulatory gene and three structural gene.
 (b) The i gene codes for inducer
 (c) The z-gene codes for β-galactosidase
 (d) A very low level of expression of lac operon has to be present in cell all the time
- (1) a, b, c, d (2) a, c, d
 (3) a, d (4) b, d
118. One of these is not associated with termination of protein synthesis :-
 (1) UAA (2) UAG
 (3) UGA (4) AUG
119. During translation ribosome moves from 5' to 3' end of m-RNA to recognise all codons. This movement of ribosome is called :-
 (1) Transformation
 (2) Translocation
 (3) Transduction
 (4) Transcription
120. A popular concept "One mutant gene-one metabolic block" given by :-
 (1) Father of modern genetics
 (2) Father of genetics
 (3) Father of experimental genetics
 (4) Father of human genetics

121. A eukaryotic gene was introduced into bacteria along with suitable promoter but it was found that this gene did not form similar protein in bacteria. It is due to the :-

- (1) Absence of RNA polymerase.
- (2) Absence of origin of replication.
- (3) Absence of splicing mechanism.
- (4) Absence of 80s ribosomes in bacteria.

122. Which of the following is a Corepressor of tryptophan operon.

- (1) Lactose
- (2) Tryptophan
- (3) Chorismic acid
- (4) Permease

123. A gene of operon which forms the repressor protein is

- (1) Operator
- (2) Promoter
- (3) Regulator
- (4) Structural

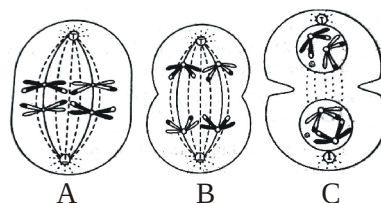
124. In lac operon genes a, i, y and z code for

- (1) Repressor protein, permease, β -galactosidase
- (2) Transacetylase, repressor protein, permease, β -galactosidase
- (3) Transacetylase, permease, β -galactosidase, repressor protein
- (4) Permease, transacetylase, repressor protein, β -galactosidase

125. *Escherichia coli* growing on medium having glucose is transferred to lactose containing medium. Which change occurs?

- (1) Lac operon is induced
- (2) Lac operon is suppressed
- (3) All operons are induced
- (4) The bacterium stops dividing

126. Identify the given figures showing meiotic phases and select the correct option.



A B C

- | | | |
|------------------|-------------|--------------|
| (1) Metaphase | Anaphase | Telophase |
| (2) Metaphase-I | Anaphase-I | Telophase-I |
| (3) Metaphase-II | Anaphase-II | Telophase-II |
| (4) Anaphase-I | Metaphase-I | Telophase-I |

127. Diakinesis is marked by

- (1) terminalisation of chiasmata
- (2) degeneration of nucleolus
- (3) fully condensed chromosomes
- (4) All of the above

128. Match the following columns.

Column I	Column II
A. Terminalisation	1. Metaphase-I
B. Histone synthesis	2. Anaphase-I
C. Disjunction	3. Diakinesis
D. Interkinesis	4. Meiosis-II
E. Double equatorial plate	5. S-phase

Codes

A	B	C	D	E
(1) 5	2	1	3	4
(2) 3	5	4	2	1
(3) 2	5	4	1	3
(4) 3	5	2	4	1

129. Crossing over occurs during :-

- (1) Diplotene
- (2) Pachytene
- (3) Zygotene
- (4) Leptotene

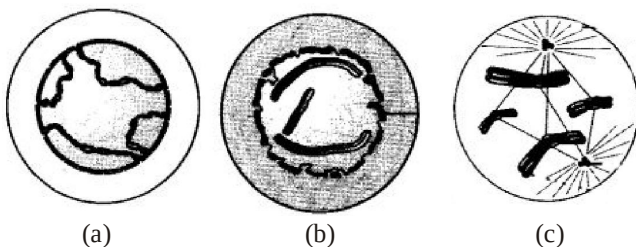
130. In Meiosis the homologous chromosomes are separated during :-

- (1) Metaphase I
- (2) Prophase II
- (3) Anaphase I
- (4) Anaphase II

131. Chromosome number is reduced to half in :-

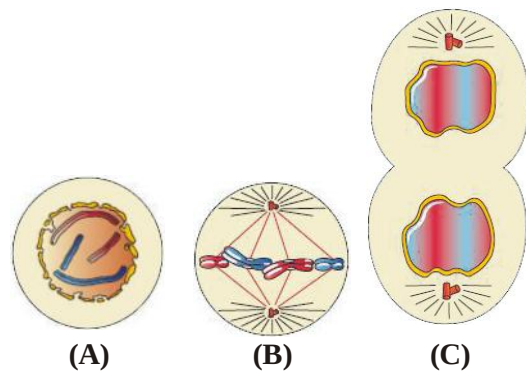
- (1) Equational division
- (2) Meiosis II
- (3) Homotypic division
- (4) Meiosis I

132. Recognise the figure and find out the correct matching :-



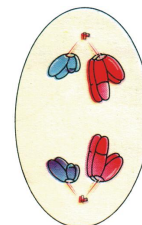
- (1) a–Early prophase, b–late prophase, c–transition to metaphase
- (2) b–Early prophase, c–late prophase, a–transition to metaphase
- (3) c–Early prophase, a–late prophase, b–transition to metaphase
- (4) b–Early prophase, a–late prophase, c–transition to metaphase

133. The figures below show 3 phases of mitosis select the option giving correct identification together with the correct event ?



- (1) C-Telophase-nuclear envelope assembles around the chromosome clusters
- (2) B-Anaphase-segregation of homologous chromosome
- (3) A-Prophase-chromosomes get fully condensed
- (4) C-Metaphase-condensation of chromatin to form chromosome

134. Identify the below stage with their respective event :-



- (1) Anaphase-II Splitting of the centromere
- (2) Anaphase-I Homologous chromosome separate
- (3) Anaphase-I Splitting of the centromere
- (4) Metaphase-I Bivalent chromosomes align on the equatorial plate.

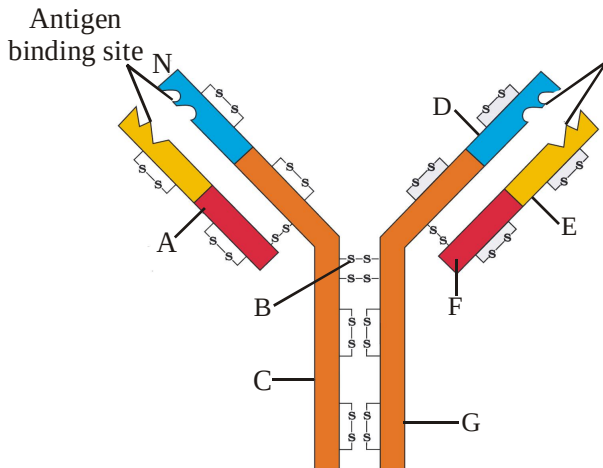
135. Bouquet stage can be observed in :-

- (1) Leptotene
- (2) Zygotene
- (3) Pachytene
- (4) Diplotene

136. The cells which in reality release the antibodies are:-

- (1) Plasma cells
- (2) Helper -T cells
- (3) Cytotoxic T-cells
- (4) Memory cells

137. Identify A, B, C, D, E, F and G as labelled in the diagram:-



Options:-

	A	B	C	D	E	F	G
(1)	Light chain	Disulphide bond	Light chain	Variable region of heavy chain	Variable region of light chain	Constant region of light chain	Constant region of heavy chain
(2)	Light chain	Disulphide bond	Light chain	Constant region of heavy chain	Constant region of light chain	Variable region of light chain	Variable region of heavy chain
(3)	Light chain	Disulphide bond	Light chain	Constant region of heavy chain	Variable region of light chain	Constant region of light chain	Variable region of heavy chain
(4)	Light chain	Disulphide bond	Heavy chain	Variable region of heavy chain	Variable region of light chain	Constant region of light chain	Constant region of heavy chain

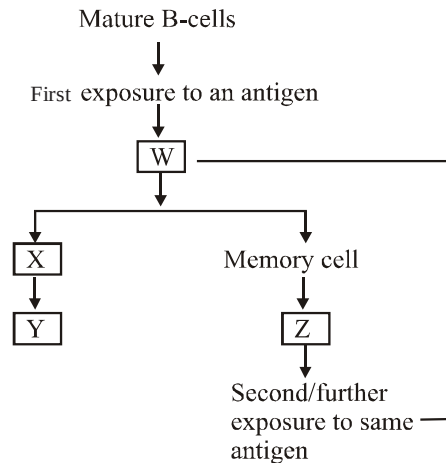
138. T-lymphocytes:-

- (1) Originates in thymus
- (2) Mature in bone marrow
- (3) Mature in thymus
- (4) Play a central role in humoral immunity

139. The largest antibody is:-

- (1) IgM
- (2) IgG
- (3) IgD
- (4) IgA

140. Identify W, X, Y and Z in the flow chart showing fate of B-cells in humoral immunity:-



Options	W	X	Y	Z
(1)	Clones of B-cells	Memory cells	Plasma cells	Memory cells degenerate soon
(2)	Plasma cells	Memory cells	Antibody synthesis by memory cells	Memory cells stored in lymph nodes
(3)	T-cells	Memory cells	Antibody synthesis	Memory cells stored in spleen
(4)	Clones of B-cells	Plasma cells	Antibody synthesis	Memory cells stored in spleen and lymph nodes

141. Find out which of the following statements are true (T)/false (F) and choose the correct option:-

- (I) MALT is absent in lining of respiratory, digestive and urogenital tracts
- (II) Antigens trapped in the lymph nodes are responsible for the activation of lymphocytes present there
- (III) Both bone-marrow and thymus provide micro-environments for the development and maturation of T-lymphocytes
- (IV) Lymph nodes serve to trap the micro-organisms or other antigens, which happen to get into the lymph and tissue fluid

Options:-

	I	II	III	IV
(1)	T	T	T	F
(2)	F	T	F	T
(3)	T	T	F	F
(4)	F	T	T	T

142. When a host is exposed to antigens, which may be in the form of living or dead microbes or other proteins, antibodies are produced in the host body. This type of immunity is called :-
 (1) Active immunity
 (2) Passive immunity
 (3) Auto immunity
 (4) Allergy
143. Rejection of transplanted organ is prevented by regular use of :-
 (1) Cyclosporin (2) Calcitonin
 (3) Thrombin (4) Pyroxin
144. Element required to maintain structure of immunoglobulin is :-
 (1) S (2) Ca (3) P (4) Fe
145. It helps in differentiation of cells of immune system :
 (1) Cortisol (2) Thyroxine
 (3) Thymosin (4) Steroid
146. Which one acts as physiological barrier to entry of micro-organisms in human body ?
 (1) Monocytes
 (2) Skin
 (3) Epithelium of urinogenital tract
 (4) Tears
147. Obstacle to large scale transplantation of organs is:
 (1) Insufficiency of organ donors
 (2) Immunological rejection of foreign bodies
 (3) Religious or ethnic considerations
 (4) Lack of effective surgical techniques

148. Ability of the host to fight the disease causing organisms conferred by the immune system is called as.
 (1) Immunology (2) Immunity
 (3) Immunotherapy (4) 1 & 2 both
149. Certain types of leukocytes of our body are phagocytes except?
 (1) Monocyte
 (2) PMNL-neutrophil
 (3) Large granular lymphocyte
 (4) Macrophage
150. Match the columns with regard to innate immunity:-

	Column-I		Column-II
(A)	Gastric juice, saliva Tears & sweat	(i)	Cytokine barrier
(B)	Interferons	(ii)	Physical barrier
(C)	Mucosa & skin	(iii)	Cellular barrier
(D)	PMNL-neutrophils monocyte, N-K cells	(iv)	Physiological barrier

- (1) A-ii, B-i, C-iv, D-iii
 (2) A-iii, B-i, C-ii, D-iv
 (3) A-iv, B-i, C-ii, D-iii
 (4) A-iv, B-i, C-iii, D-ii
151. The activated lymphocyte that neither produce antibody nor kill the infected cells developed in to?
 (1) N-K cell (2) T_K-cells
 (3) Memory cells (4) 1 & 3 both
152. Select the correct statements.
 (a) Injection of snake antivenom against snake bite is an example of active immunisation
 (b) If due to some reason B & T lymphocyte are damaged, the body will not produce antibody against a pathogen
 (c) Injection of dead/inactivated pathogen causes passive immunity
 (d) Colostrum contain maximum I_gA
- How many statements are correct ?
 (1) One (2) Two
 (3) Three (4) Four

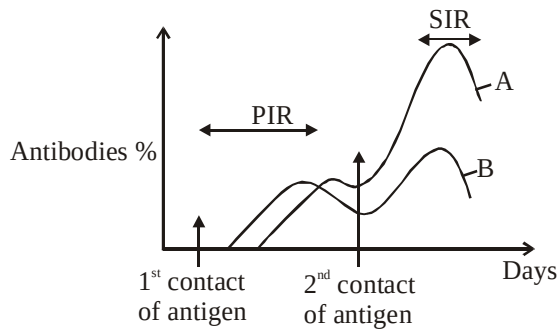
153. After maturation the lymphocytes migrate to?
- Thymus
 - Spleen
 - Secondary lymphoid organs
 - 2 & 3 both
154. Which group of drugs quickly reduce the symptoms of allergy :-
- Histamine, Adrenaline, Steroids
 - Histamine, Acetylcholine, Steroids
 - Antihistaminics, Adrenaline, Steroids
 - Antihistaminics, cyclosporin, steroids
155. Rejection of non self organs can be prevented by use of ____ ?
- cellular immunity
 - T-lymphocyte
 - 1 & 2 both
 - Immuno suppressant
156. Which of the following is incorrect about Natural Killer Cells ?
- Contain perforin protein
 - Large agranular lymphocyte
 - Non specific in function
 - Produced in bone marrow
157. Tissue graft in between the genetically dissimilar individual of the same species is :-
- Autograft
 - Isograft
 - Xenograft
 - Allograft
158. Transplantation of tissues/organs to save certain patients often fails due to rejection of such tissues/organs by the patient. Which type of immune response is responsible for such rejections ?
- auto-immune response
 - humoral immune response
 - physiological immune response
 - cell-mediated immune response
159. Anti venom against snake poison contains :
- Antigens
 - Antigen-antibody complex
 - Antibodies
 - Enzymes

160. Which of the following statement is false ?
- Virus infected cells secrete interferons
 - Interferons protect non infected cells from further viral infection
 - Innate immunity is specific type of defence, that is present at the time of birth
 - The overall ability of the host to fight the disease causing organisms, conferred by the immune system is called immunity
161. Which of the following components does not participate in innate immunity ?
- Neutrophils
 - Macrophages
 - B-lymphocytes
 - Natural killer cells
162. Short-lived immunity acquired from mother to foetus across placenta and through mother's milk to infant is respectively due to :-
- IgG, IgM
 - IgA, IgG
 - IgG, IgA
 - IgG, IgD
163. Which of the following options give correct categorisation of antibody quantity and functions. :-

	Column-I Antibody	Column-II Quantity (%)	Column-III Function
1	IgA	a 5-10%	i Stimulation of mast cells or basophils
2	IgE	b 1-3%	ii Activation of B-cell
3	IgD	c 0.05%	iii Protects blood stream
4	IgM	d 10%	iv Protection of body surfaces

- 1=a-iii, 2=c-i, 3=b-ii, 4=d-iv
- 1=a-i, 2=b-ii, 3=c-iii, 4=d-iv
- 1=d-iv, 2=c-i, 3=a-iii, 4=b-ii
- 1=d-iv, 2=c-i, 3=b-ii, 4=a-iii

164.



Labelled B is :-

- (1) IgM (2) IgG
(3) IgD (4) IgA

165. Match the column-I and II and choose the correct combination from the options given :

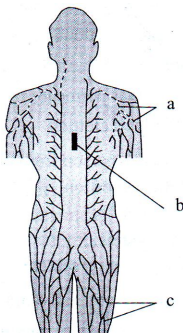
Column-I	Column-II
(a) Natural active	(p) Immunity device opted by heredity
(b) Artificial passive	(q) From mother foetus through placenta
(c) Artificial active	(r) Injection of antiserum
(d) Natural passive	(s) Fighting infection naturally
	(t) Induced vaccination

- (1) a-s, b-t, c-q, d-r
(2) a-p, b-q, c-r, d-t
(3) a-t, b-s, c-r, d-p
(4) a-s, b-r, c-t, d-q

166. Shape of antibody is like :

- (1) X (2) Z (3) Y (4) O

167. Recognise the figure and find out the correct matching



- (1) a-lymphatic vessels, b-thymus, c-lymph nodes
(2) c-lymphatic vessels, a-thymus, b-lymph nodes
(3) b-lymphatic vessels, c-thymus, a-lymph nodes
(4) c-lymphatic vessels, b-thymus, a-lymph nodes

168. MALT constitutes about _____ percent of the lymphoid tissue in human body.

- (1) 20% (2) 70%
(3) 10% (4) 50%

169. Peyer's patches are found in :

- (1) Small intestine
(2) Large intestine
(3) Stomach
(4) Vermiform appendix

170. Immature lymphocyte differentiates into antigen sensitive lymphocytes in the :-

- (1) Primary lymphoid organs
(2) Secondary lymphoid organs
(3) Both 1 and 2
(4) None of these

171. In stomach which one comes :

- (1) Bolus
(2) Chyme
(3) Chyle
(4) All

172. The structural and functional unit of liver is :

- (1) Glisson's capsule
(2) Hepatic lobules
(3) Hepatic vein
(4) Hepatic cords

173. Which of the following muscular sphincter regulates the opening of oesophagus into the stomach ?

- (1) Pyloric sphincter
(2) Gastro-oesophageal sphincter
(3) Sphincter of Boyden
(4) Sphincter of oddi

174. Match the following :

Column I	Column II (pH)
I. Acid in stomach	(a) 6.8
II. Salivary amylase	(b) 7.8
III. Pancreatic juice	(c) 1.8
(1) I = a, II = b, III = c	
(2) I = b, II = a, III = c	
(3) I = c, II = a, III = b	
(4) I = c, II = b, III = a	

- | | |
|---|---|
| <p>175. Which one of the following is not a cell of gastric glands :</p> <p>(1) Mucus neck cells
(2) Paneth cells
(3) Peptic cells
(4) Parietal cells</p> <p>176. Bile is formed in :</p> <p>(1) Gall bladder
(2) Liver
(3) Spleen
(4) Blood</p> <p>177. Emulsification of fat will not occur in the absence of :</p> <p>(1) Lipase
(2) Amylase
(3) Bile salts
(4) Pancreatic juice</p> | <p>178. The distal part of the stomach that opens into duodenum is called :</p> <p>(1) Fundus
(2) Pylorus
(3) Omentum
(4) Cardiac</p> <p>179. Which part of alimentary canal does not secrete any digestive enzyme ?</p> <p>(1) Mouth
(2) Stomach
(3) Duodenum
(4) Oesophagus</p> <p>180. Find the odd one out</p> <p>(1) duodenum
(2) jejunum
(3) ileum
(4) caecum</p> |
|---|---|



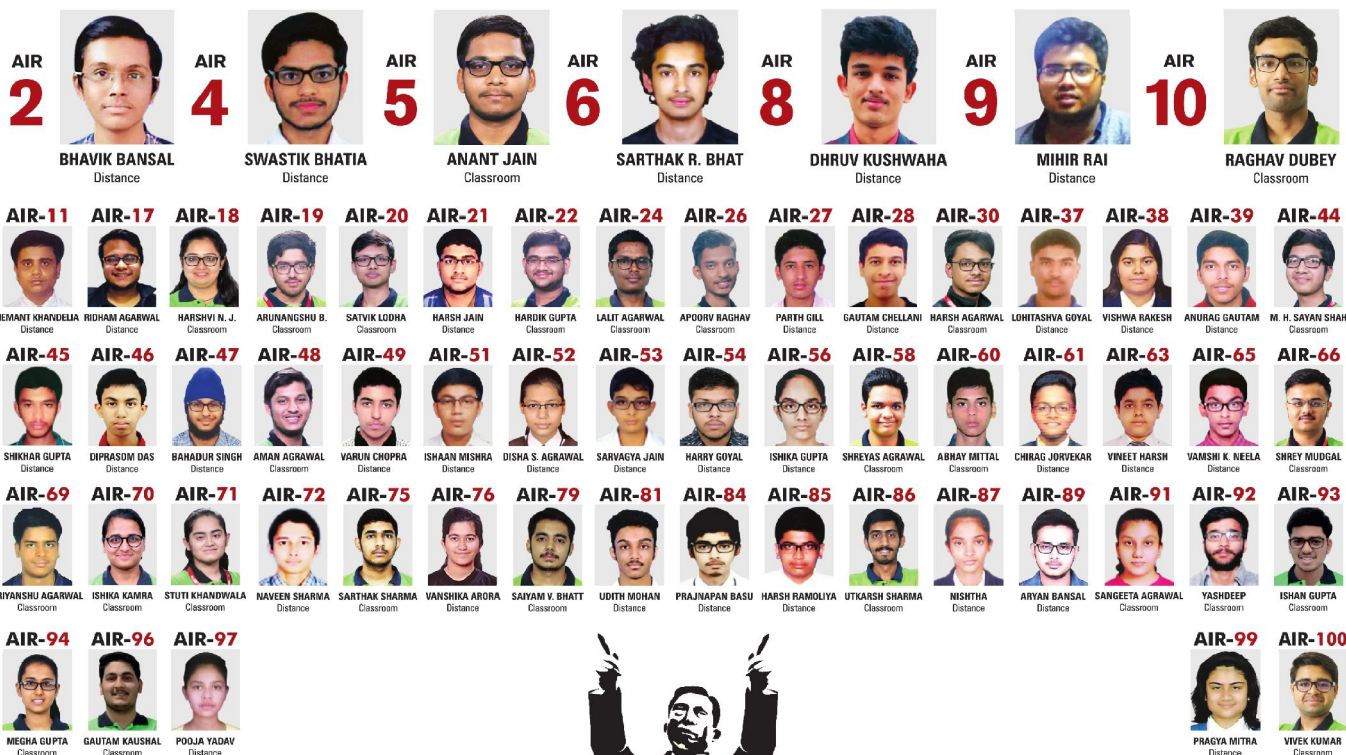
Space For Rough Work

ALLEN है... तो मुमकिन है

 **61 in Top 100 All India Ranks**

95102*

NALIN KHANDELWAL
Classroom



A - New Products Committee] and J. A. Lee

 **ALLEN**
CAREER INSTITUTE
KOTA (RAJASTHAN)

Heartiest
Congratulations!